

Rotor Sweep

Q-1240-032

ITP step #: 6000-2

Revision # 01

Project: GMS

Unit : 4

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Number of poles: 48

Readings in : in

When reading is taken, the pole is located upstream.

Pole # at 0°	Top	Bottom	Avg.
2	0.747	0.726	0.736
4	0.748	0.722	0.735
6	0.750	0.737	0.743
8	0.753	0.739	0.746
10	0.742	0.731	0.737
12	0.738	0.729	0.733
14	0.740	0.729	0.735
16	0.731	0.724	0.728
18	0.729	0.731	0.730
20	0.741	0.739	0.740
22	0.746	0.740	0.743
24	0.743	0.740	0.742

Pole # at 0°	Top	Bottom	Avg.
26	0.740	0.723	0.732
28	0.734	0.725	0.729
30	0.736	0.723	0.730
32	0.730	0.728	0.729
34	0.738	0.722	0.730
36	0.741	0.730	0.736
38	0.749	0.742	0.745
40	0.747	0.740	0.744
42	0.736	0.724	0.730
44	0.740	0.739	0.740
46	0.745	0.745	0.745
48	0.750	0.741	0.745

*** 0 Degree position taken at upstream on stator (punching between slot #612 and slot #1)



Notes : Readings taken at 150mm (6") from end of pole iron.

See page 2 for dial indicator reference and results

NCR #: N/A

Measurement Instruments	T°	Measured by:	Ross Carter	dd-mmm-yyyy
Slab Gauge	Before: <u>22 C</u>		Greg Gacek	April 29-30th 2011
Outside Micrometer 0-1"	After: <u>22 C</u>	Andritz Rep:	Jean-Luc Cote <i>JC</i>	30-Apr-2011
#951069A		Cust. Repres.	Bruce Malone <i>BM</i>	30-Apr-2011

Rotor Sweep

Q-1240-032

ITP step #: 6000-2

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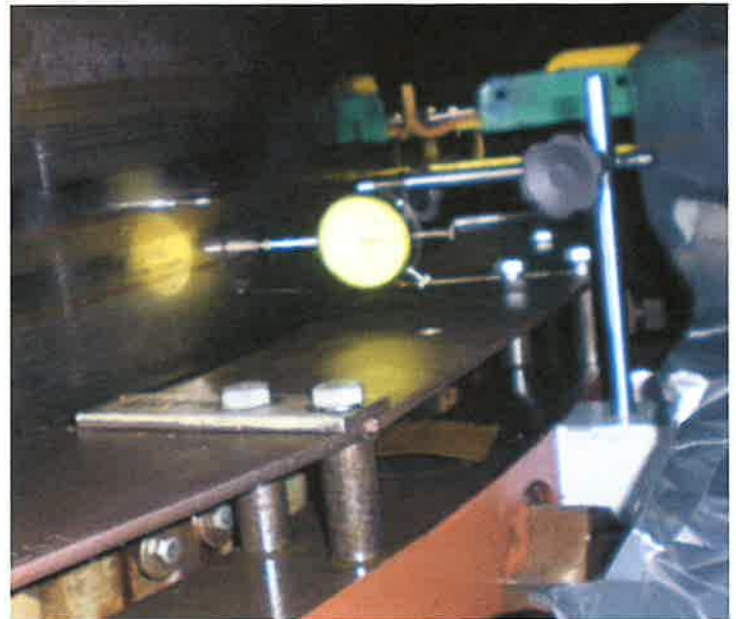
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Dial indicator used for ROTOR sweep

Readings in : in

Dial indicator at thrust ring/block			
Pole	Upstream	Pole	Upstream
2	0.001	26	0.002
4	0.001	28	0.002
6	0.001	30	0.001
8	0.001	32	0.001
10	0.002	34	0.001
12	0.002	36	0.001
14	0.002	38	0.001
16	0.002	40	0.001
18	0.002	42	0.002
20	0.002	44	0.002
22	0.002	46	0.001
24	0.001	48	0.000



Rotor		
	Tolerance	Result
Concentricity ⁽¹⁾	0.014	0.004
Circularity ⁽²⁾	0.092	0.019

NCR #: N/A

Measurement Instruments	T°	Measured by:	Ross Carter	dd-mmm-yyyy
Slab Gauge	Before: <u>22 C</u>		Greg Gacek	April 29-30th 2011
Outside Micrometer 0-1"	After: <u>22 C</u>	Andritz Rep:	Jean-Luc Cote <i>JLC</i>	30-Apr-2011
#951069A		Cust. Repres.	Bruce Malone <i>BM</i>	30-Apr-2011

Installation and Commissioning Quality Control Record

Reference elevation difference between rotor and stator **Q-GMS-6301-050**

 ITP step #: 6000-03 & 04

 Revision # 00

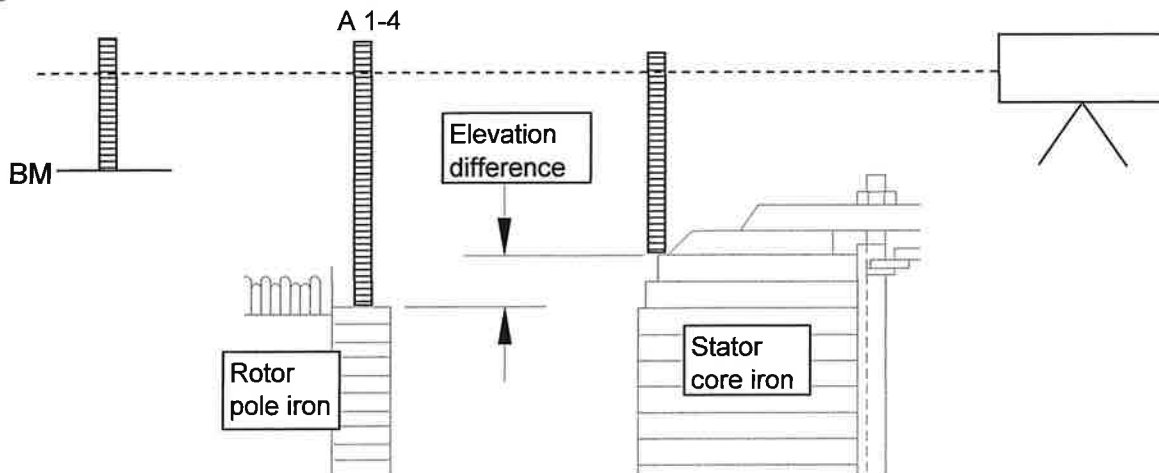
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 Readings in : inches

- ☒ At disassembly - Elevation taken below the rotor
☐ At reassembly



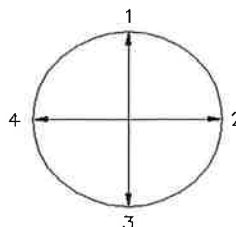
Reading must be taken while rotor installed.

ELEVATIONS

Bench mark elevation: 0.000 The BM act as the reference

Stator teeth next to slot # 3o'clock : 72.613 The stator slots were not identified. The teeth used was marked and is located at 3o'clock

Rotor pole	A
48	72.820
12	72.673
24	72.692
36	72.821



NCR #: _____

Measurement Instruments	T°	Measured by:	Greg Gacek	dd-mmm-yyyy
N3 #15718	Before: 20C		Ross Carter	28-Apr-2011
Whyteface #5080	After: 20C	Andritz Rep.:	F. Sergerie, JL Cote	28-Apr-2011
		Cust. Repres.:	BRUCE MALONE BM	MAY 04/11

General Inspection Report

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ITP step #: 6000-5

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Equipment : Pole Keys grease samples for analysis

No grease could be observed from the bottom of the rotor pole keys



As a result, dust power was collected using a knife and scraping the surface of the keys



NCR #: _____

Measurement Instruments	T°	Measured by:	Jean-Luc Cote	dd-mmm-yyyy
Visual	Before: N/A			
	After: N/A	ANDRITZ Rep.:	Jean-Luc Cote	1-May-2011
		Cust. Repres.	Bruce Malone	<i>B MS MAY 04/11</i>

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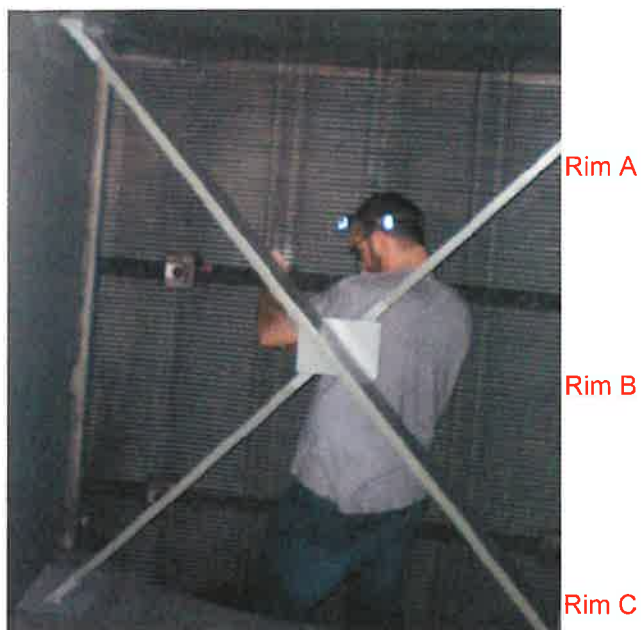
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Equipment : Relative Tangential & Radial Rotor Rim Alignment



Method of Measurement for Tangential Offset



Method of Measurement for Radial Offset

*** See page 2 & 3 for data

NCR #: _____

Measurement Instruments	T°	Measured by:	Greg Gacek	dd-mmm-yyyy
Straight Edge	Before: 22 C		Ross Carter	28-Apr-2011
Feeler Gauges	After: 22 C	ANDRITZ Rep.:	JL Cote, F.Sergerie	29-Apr-2011
		Cust. Repres.	BRUCE MALONE BR	

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Equipment : Relative Tangential Rotor Rim Alignment

Tangential Rim Alignment		
Pocket #	Rim A to Rim B	Rim B to Rim C
1	Rim A is 0.015" further clockwise to Rim B	Rim B is 0.034" further clockwise to Rim C
2	Rim A is 0.020" further clockwise to Rim B	Rim B is 0.054" further clockwise to Rim C
3	Rim A is 0.017" further clockwise to Rim B	Rim B is 0.036" further clockwise to Rim C
4	Rim A is 0.025" further clockwise to Rim B	Rim B is 0.040" further clockwise to Rim C
5	Rim A is 0.015" further clockwise to Rim B	Rim B is 0.031" further clockwise to Rim C
6	Rim A is 0.023" further clockwise to Rim B	Rim B is 0.027" further clockwise to Rim C
7	Rim A is 0.014" further clockwise to Rim B	Rim B is 0.027" further clockwise to Rim C
8	Rim A is 0.025" further clockwise to Rim B	Rim B is 0.015" further clockwise to Rim C
9	Rim A is 0.020" further clockwise to Rim B	Rim B is 0.045" further clockwise to Rim C
10	Rim A is 0.034" further clockwise to Rim B	Rim B is 0.025" further clockwise to Rim C
11	Rim A is 0.038" further clockwise to Rim B	Rim B is 0.015" further clockwise to Rim C
12	Rim A is 0.020" further clockwise to Rim B	Rim B is 0.028" further clockwise to Rim C
13	Rim A is 0.052" further clockwise to Rim B	Rim B is 0.043" further clockwise to Rim C
14	Rim A is 0.038" further clockwise to Rim B	Rim B is 0.031" further clockwise to Rim C
15	Rim A is 0.048" further clockwise to Rim B	Rim B is 0.015" further clockwise to Rim C
16	Rim A is 0.025" further clockwise to Rim B	Rim B is 0.022" further clockwise to Rim C
17	Rim A is 0.048" further clockwise to Rim B	Rim B is 0.027" further clockwise to Rim C
18	Rim A is 0.042" further clockwise to Rim B	Rim B is 0.027" further clockwise to Rim C
19	Rim A is 0.033" further clockwise to Rim B	Rim B is 0.042" further clockwise to Rim C
20	Rim A is 0.036" further clockwise to Rim B	Rim B is 0.030" further clockwise to Rim C

*** See page 1 for identification & pictures

NCR #: _____

Measurement Instruments	T°	Measured by:	Greg Gacek	dd-mmm-yyyy
Straight Edge	Before: <u>22 C</u>		Ross Carter	<u>28-Apr-2011</u>
Feeler Gauges	After: <u>22 C</u>	ANDRITZ Rep.:	JL Cote, F. Sergerie	<u>29-Apr-2011</u>
		Cust. Repres:	<u>BRUCE MALONE RM</u>	

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Equipment : Relative Radial Rotor Rim Alignment

Radial Rim Alignment		
Pocket #	Rim A to Rim B	Rim B to Rim C
1	Rim A is even with Rim B	Rim B is even with Rim C
2	Rim A is even with Rim B	Rim B is even with Rim C
3	Rim A is even with Rim B	Rim B is even with Rim C
4	Rim A is even with Rim B	Rim B is even with Rim C
5	Rim A is even with Rim B	Rim B is even with Rim C
6	Rim A is even with Rim B	Rim B is even with Rim C
7	Rim A is even with Rim B	Rim B is even with Rim C
8	Rim A is even with Rim B	Rim B is even with Rim C
9	Rim A is even with Rim B	Rim B is even with Rim C
10	Rim A is even with Rim B	Rim B is even with Rim C
11	Rim A is even with Rim B	Rim B is even with Rim C
12	Rim A is even with Rim B	Rim B is even with Rim C
13	Rim A is even with Rim B	Rim B is even with Rim C
14	Rim A is even with Rim B	Rim B is even with Rim C
15	Rim A is even with Rim B	Rim B is even with Rim C
16	Rim A is 0.012" closer to center than Rim B	Rim B is even with Rim C
17	Rim A is 0.011" closer to center than Rim B	Rim B is even with Rim C
18	Rim A is even with Rim B	Rim B is even with Rim C
19	Rim A is even with Rim B	Rim B is even with Rim C
20	Rim A is even with Rim B	Rim B is even with Rim C

*** See page 1 for identification & pictures

*** "Even" implies that the straight edge would not catch any defect with sliding up and down from one rim to the other

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Greg Gacek</u>	dd-mmm-yyyy
<u>Straight Edge</u>	Before: <u>22 C</u>	<u>Ross Carter</u>	<u>28-Apr-2011</u>
<u>Feeler Gauges</u>	After: <u>22 C</u>	ANDRITZ Rep.: <u>JL Cote, F. Sergerie</u>	<u>29-Apr-2011</u>
		Cust. Repres. <u>BRUCE MALONE Rm</u>	

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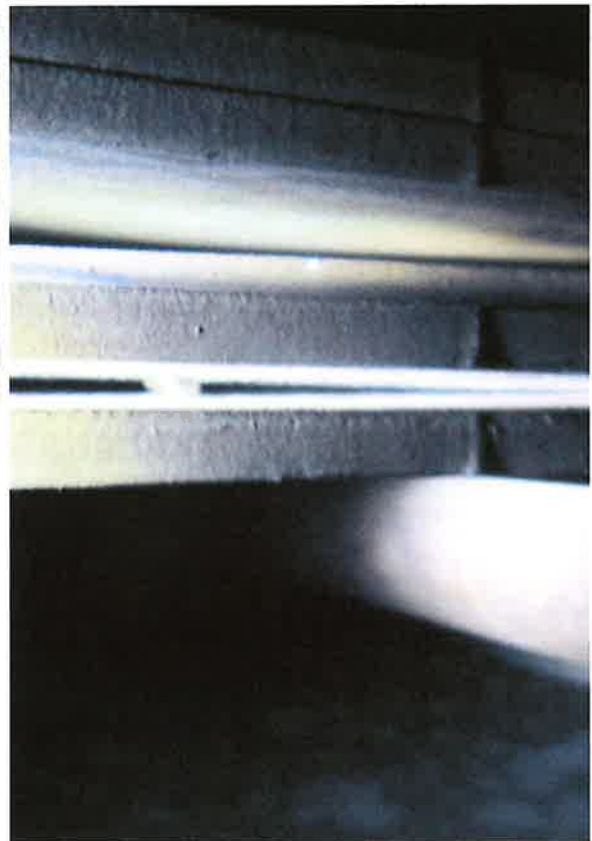
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 Equipment : Rotor rim visual inspection

Nothing major to report on the rotor rim visual inspection.

In each rim section, the rim plates are well aligned.

On metal particle was found between two rim plates in pocket #20, at the bottom of the top section. It must have been there since the assembly because it is about 150mm deep into the rim and the bottom plate bent because of it.



NCR #: _____

Measurement Instruments	T°	Measured by:	Greg Gacek	dd-mmm-yyyy
Visual	Before: 22 C		Ross Carter	28-Apr-2011
	After: 22 C	ANDRITZ Rep.:	JL Cote, F.Sergerie	29-Apr-2011
		Cust. Repres	Bouice Malone RM	MAY 04/11

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Equipment : Rotor rim fretting

Some fretting ares were found in the rotor.

In all the pockets, the fretting can be found next to the rim, on the very top, where it meets with the rotor spider top plate.



NCR #: _____

Measurement Instruments	T°	Measured by:	Greg Gacek	dd-mmm-yyyy
Visual	Before: 22 C		Ross Carter	28-Apr-2011
	After: 22 C	ANDRITZ Rep.:	JL Cote, F.Sergerie <i>OK</i>	29-Apr-2011
		Cust. Repres.	<i>BAUCE MALONE BR MAY 04/11</i>	

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Equipment : Rotor lead visual inspection

A burned area was found on one of the rotor leads that goes from the pole to the collector.

It is located on the horizontal section, on the rotor floor, inside the rotor central hub.



NCR #: _____

Measurement Instruments	T°	Measured by:	Greg Gacek	dd-mmm-yyyy
Visual	Before: 22 C		Ross Carter	28-Apr-2011
	After: 22 C	ANDRITZ Rep.:	JL Cote, F.Sergerie	29-Apr-2011
		Cust. Repres.	SAUCE MALONE RM	MAY 04/11

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 Equipment : Visual inspection of the welds

Some cracks in the welds were found in the rotor.

1 - In rotor pocket #9, at the bottom right corner, between the plate and the spider arm.

This crack is about 300mm long.



NCR #: _____

Measurement Instruments	T°	Measured by:	<u>Greg Gacek</u>	dd-mmm-yyyy
<u>Visual</u>	Before: <u>22 C</u>		<u>Ross Carter</u>	<u>28-Apr-2011</u>
	After: <u>22 C</u>	ANDRITZ Rep.:	<u>JL Cote, F.Sergerie</u>	<u>29-Apr-2011</u>
		Cust. Repres.	<u>BRUCE MALONE</u>	<u>29 MAY 04/11</u>

General Inspection Report

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ITP step #: 6000-06

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Equipment : Visual inspection of the welds

Some cracks in the welds were found in the rotor.

2 - In rotor pocket #9, at the bottom right corner, between the rotor floor and a balance weight
This crack is about 100mm long.



NCR #: _____

Measurement Instruments	T°	Measured by:	Greg Gacek	dd-mmm-yyyy
Visual	Before: 22 C		Ross Carter	28-Apr-2011
	After: 22 C	ANDRITZ Rep.:	JL Cote, F.Sergerie	29-Apr-2011
		Cust. Repres.	BRUCE MALONE RM	MAY 04/11

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 ITP step #: 6000-06

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 Equipment : Visual inspection of the welds

Some cracks in the welds were found in the rotor.

3 - In rotor pocket #11, at the top, left hand side, approximately to the middle, there appears to be a crack of approximately 1/2" long.



NCR #: _____

Measurement Instruments	T°	Measured by:	Greg Gacek	dd-mmm-yyyy
Visual	Before: 22 C		Ross Carter	28-Apr-2011
	After: 22 C	ANDRITZ Rep.:	JL Cote, F.Sergerie	29-Apr-2011
		Cust. Repres:	DAVID MALONE	MAY 04/11

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Equipment : Visual inspection of the welds

Some cracks in the welds were found in the rotor.

4 - In rotor pocket #11, at the bottom right corner on right hand side, there is presence considerable porosities into the weld.



NCR #: _____

Measurement Instruments	T°	Measured by:	Greg Gacek	dd-mmm-yyyy
Visual	Before: 22 C		Ross Carter	28-Apr-2011
	After: 22 C	ANDRITZ Rep.:	JL Cote, F. Sergerie	29-Apr-2011
		Cust. Repres.	BRUCE MALONE Sm	MAY 04/11

ITP STEP 6000-7

Unit G4 Preliminary Report

Rotor Rim Shrink Speed Measurement**Equipment:**

Bently Nevada 3300 8 mm displacement proximity sensors

ACR Owl – 400 DC voltage data loggers

Battery packs to power proximitors – custom

Brackets to hold sensor – custom made at site

Brackets to hold sensor driver, data logger and battery pack – custom made at site

Procedure:

Sensor Mounting – Brackets were already in place from a previous test by BCH in 2006, they were strengthened by welding an additional piece of angle on the inner side. A steel target was also welded to the rim for a target see Fig. 1.0. The Bently Nevada 3300 sensors are calibrated from the factory using 4140 stainless steel targets but I have found that a carbon steel target will meet the calibration requirements for this application.

**Fig. 1.0**

GM Shrum Rim Shrink Speed Measurement

Unit G4 Preliminary Report

The driver for the 3300 proximity sensor was mounted in metal bracket above the brake track see Fig. 2.0. Each measurement point required one proximity sensor and driver along with a custom made power supply and one data logger. The battery pack and data logger were also mounted inside this bracket. All metal brackets were left in place for future use.



Fig. 2.0

GM Shrum Rim Shrink Speed Measurement

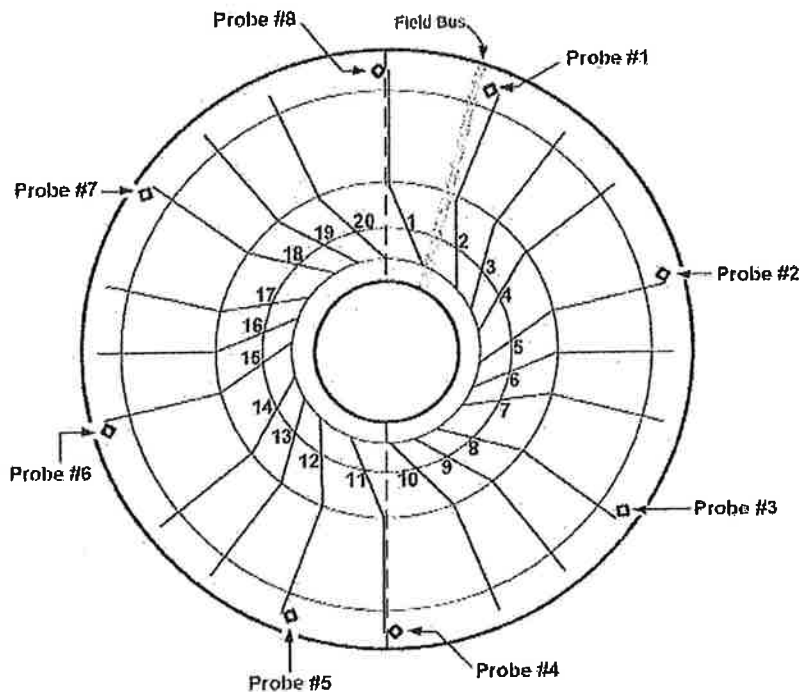
Unit G4 Preliminary Report

Measurement:

The data loggers were activated prior to releasing the unit clearance and sample automatically every 5.0 seconds. All data loggers have an internal clock and were synchronized to the power house control room. The unit was operated mechanically from a minimum speed to 100 % speed in predetermined steps and allowed to operate at each point for 10 minutes to allow the data loggers to acquire data, unit speed was hand recorded from the control room along with the time, also a file of unit speed was acquired from the unit monitoring system.

Sensor Locations:

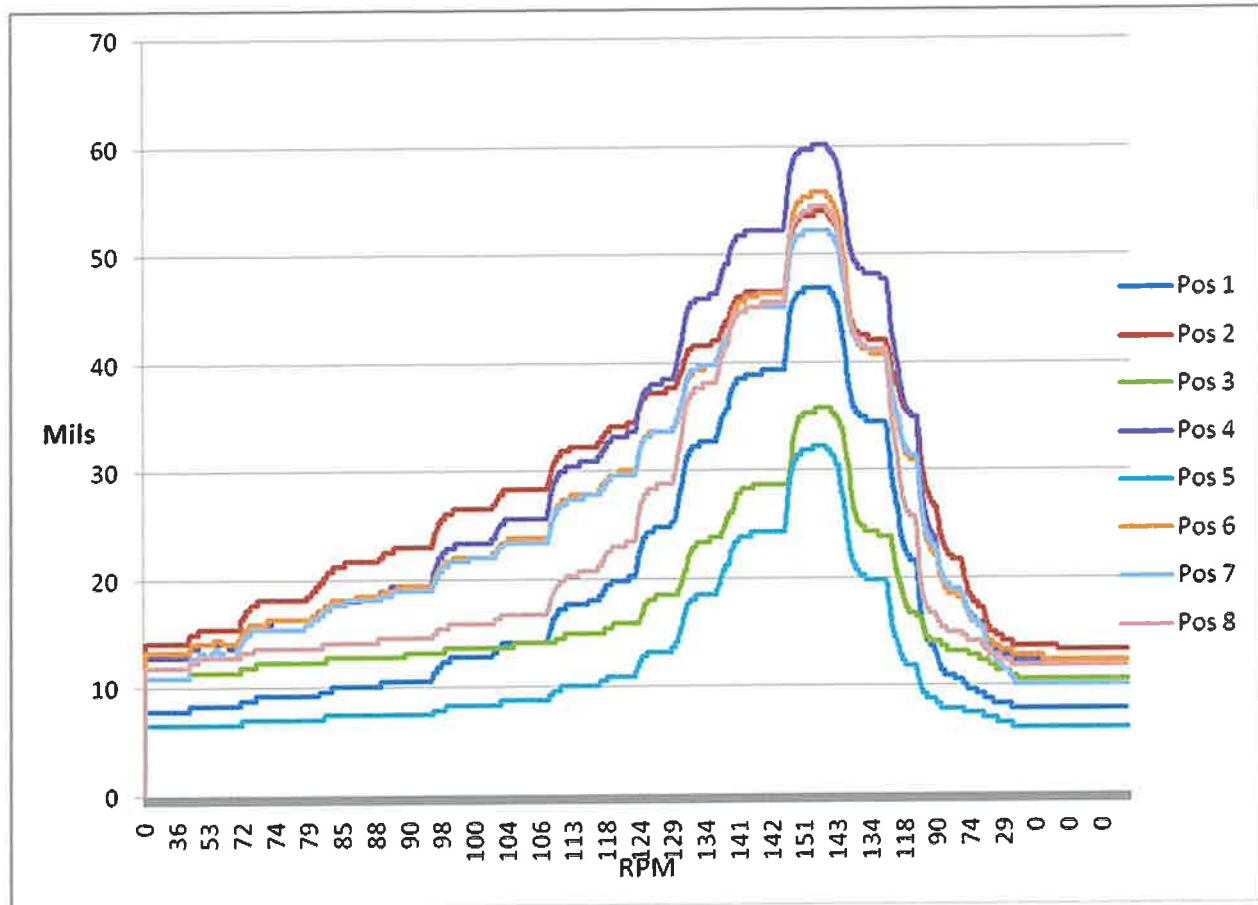
The same locations on the lower rim section were used as in the previous 2006 test.



GM Shrum Rim Shrink Speed Measurement

Unit G4 Preliminary Report

Measurement Chart:



NOTE: FOR ENGINEERING ASSESSMENT OF THESE MEASUREMENTS.
SEE ANDRITZ SUBMITTAL SP5.2.2 - G4 CONDITION
ASSESSMENT REPORT.